Predicting Truck Failures Using Machine Learning

Zoe Martinez

Dr. Eduardo Ceh-Varela

Department of Mathematical Sciences

{eduardo.ceh, zoe.martinez}@enmu.edu

Introduction

Truck failures can cause high costs, downtime, and safety risks, making early prediction important. This project uses machine learning to predict failures based on sensor data. The dataset is highly imbalanced, meaning failures are rare compared to normal cases. The goal is to identify the model that performs best under these conditions.

Dataset

★ APS Failure Dataset (UCI)
★ 60,000 samples, 170 features
★ Highly imbalanced (~11,800 normal vs 200 failures)
★ Missing values handled using median imputation

Methodology

The data was cleaned, converted to numeric form, and standardized. It was split into training and testing sets while preserving class distribution. Multiple models were trained, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Hyperparameter tuning was performed using GridSearchCV, and performance was evaluated using accuracy, precision, recall, and F1-score.

Results

XGBoost achieved the best performance with an F1-score of 0.7912. Although all models had high accuracy (~99%), this was misleading due to class imbalance. Ensemble models such as Random Forest and XGBoost performed better overall. XGBoost provided the best balance between detecting failures and avoiding false predictions. F1-score was prioritized due to class imbalance."

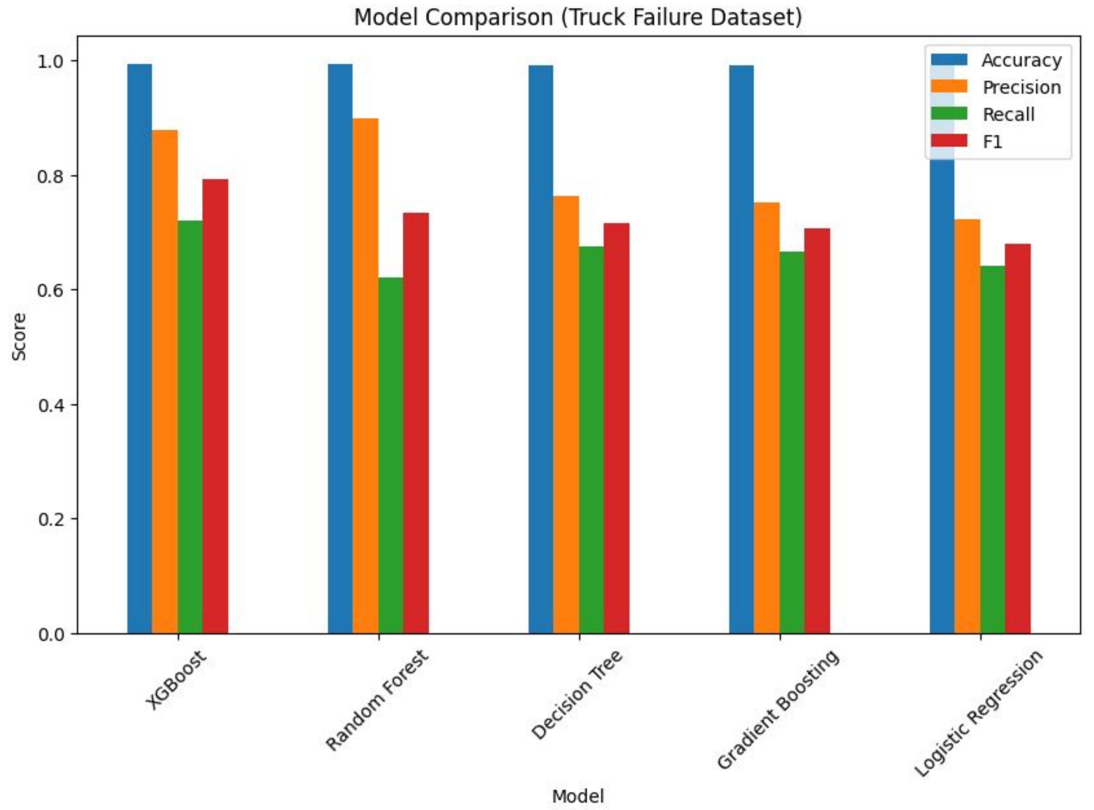
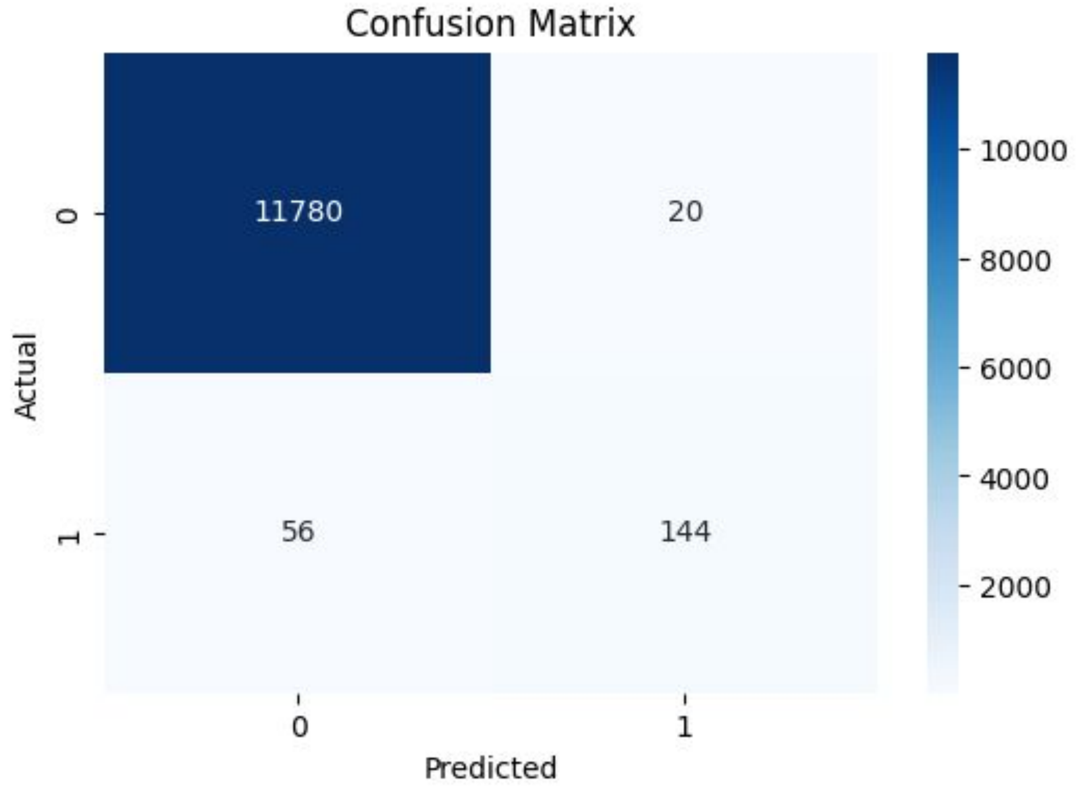


Figure 1: Model Performance Comparison

Model	F1
XGBoost	**0.7912**
Random Forest	0.7337
Decision Tree	0.7162
Gradient Boosting	0.7056
Logistic Regression	0.6790

Figure 2: Confusion Matrix (XGBoost Model)



Conclusion

XGBoost was the best-performing model due handle complex and high-dimensional data. T the importance of using F1-score instead of a imbalanced datasets. Ensemble models cons outperformed simpler models. This approach real-world predictive maintenance.

Future Work

Future work could focus on improving failure techniques like SMOTE or resampling. Additi engineering may improve model performance models could also be explored for more comp improvements could enhance real-world depl

References

APS Failure Dataset

UCI Machine Learning Repository.

https://archive.ics.uci.edu/dataset/414/ida201

XGBoost Paper (optional but strong)

Chen, T., & Guestrin, C. (2016).

XGBoost: A Scalable Tree Boosting System.